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PHYSICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Wave on String

English Medium

EXERCISE (O-1)

A. Equation of travelling wave

1. The function of x and t that does not represent a progressive wave is :-

(A) $y = 2\sin(4t - 3x)$ (B) $y = e^{(4+(4t-3x))}$ (C) $y = [4t - 3x]^{-1}$ (D) $y = [4t - 3x]$

WA0023

2. At x = 0 particle oscillate by law $y = \frac{3}{2t^2+1}$. If wave is propagating along –ve x axis with velocity 2m/s. Find equation of wave

(A) $y = \frac{3}{2\left(t-\frac{x}{2}\right)^2+1}$ (B) $y = \frac{3}{2\left(t+\frac{x}{2}\right)^2+1}$ (C) $y = \frac{3}{2\left(t-\frac{z}{2}\right)^2+1}$ (D) $y = \frac{3}{2\left(t+\frac{z}{2}\right)^2+1}$

WA0024

3. The shape of a wave propagating in the positive x or negative x- direction is given $y = \frac{1}{\sqrt{1+x^2}}$ at t = 0 and $y = \frac{1}{\sqrt{2-2x+x^2}}$ at t = 1s where x and y are in meters. The shape the wave disturbance does not change during propagation. Find the velocity of the wave.

(A) 1 m/s in positive x direction (B) 1 m/s in negative x direction

(C) $\frac{1}{2}$ m/s in positive x direction (D) $\frac{1}{2}$ m/s in negative x direction

WA0025

4. If at t = 0 a travelling wave pulse on a string is described by the function $y = \frac{10}{\left(5+x^2\right)}$, the wave function representing the pulse at time t, if the pulse is travelling along positive x axis with speed 2 ms^{-1} will be

(A) $\frac{10}{5+(x+2t)^2}$ (B) $\frac{10+2t}{5+x^2}$ (C) $\frac{10}{5+(x-2t)^2}$ (D) None

WA0088

5. A tuning fork makes 256 vibrations per second in air. When the velocity of sound is 330 m/s, then wave length of tone emitted is

(A) 0.56 m (B) 0.89 m (C) 1.11 m (D) 1.29 m

WA0089

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6. If the displacement of a particle executing SHM is given by $y = 0.30\sin(220t + 0.64)m$, then the frequency and maximum velocity of the particle is

(A) 35 Hz, 66 m/s (B) 45 Hz, 66 m/s (C) 58 Hz, 113 m/s (D) 35 Hz, 132 m/s

WA0090

7. A transverse wave on a string is described by the equation

y(x, t) = (2.20 cm) sin [(130 rad/s)t + (15rad/m)x].

(i) Find the approximate maximum transverse speed of a point on the string.

(A) 1.2 m/s (B) 1.7 m/s (C) 2.9 m/s (D) 3.4 m/s

(ii) Find the approximate maximum transverse acceleration of a point on the string.

(A) 300 m/s^2 (B) 372 m/s^2 (C) 410 m/s^2 (D) 450 m/s^2

(iii) Find the approximate speed of wave move moving along the string.

(A) 4.2 m/s (B) 5.6 m/s (C) 7.4 m/s (D) 8.7 m/s

WA0091

8. The displacement from the position of equilibrium of a point 4 cm from a source of sinusoidal oscillations is half the amplitude at the moment t = T/6 (T is the time period). Assume that the source was at mean position at t = 0. The wavelength of the running wave is

(A) 0.96 m (B) 0.48 m (C) 0.24 m (D) 0.12 m

WA0028

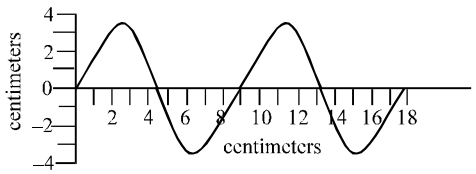
9. A sinusoidal wave travelling in the positive direction of x on a stretched string has amplitude 2.0cm, wavelength 1 m and wave velocity 5.0 m/s. At x = 0 and t = 0, it is given that displacement y = 0 and $\frac{\partial y}{\partial t} < 0$. Express the wave function correctly in the form y = f(x, t) :-

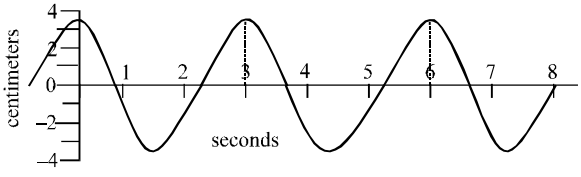
(A) y = (0.02 m) sin 2π (x–5t) (B) y = (0.02 cm) cos 2π(x–5t)

(C) y = (0.02 m) sin $2\pi\left(x - 5t + \frac{1}{4}\right)$ (D) y = (0.02 cm) cos $2\pi\left(x - 5t + \frac{1}{4}\right)$

WA0030

10. A transverse wave is travelling along a horizontal string. The first picture shows the shape of the string at an instant of time. This picture is superimposed on a coordinate system to help you make any necessary measurements. The second picture is a graph of the vertical displacement of *one* point along the string as a function of time. How far does this wave travel along the string in one second?





(A) 0.3 cm (B) 3.0 cm

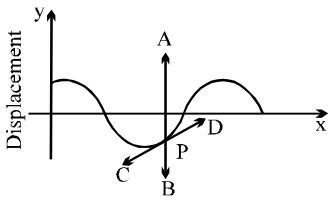
(C) 9.0 cm (D) 27 cm

WA0026

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11. The figure below shows a snap photograph of a simple harmonic progressive wave, progressing in the negative X-axis, at a given instant. The direction of the velocity of the particle at the stage P on the figure is best represented by the arrow.



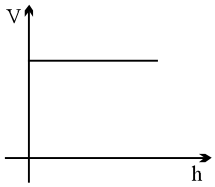
(A) $\vec{PA}$ (B) $\vec{PB}$

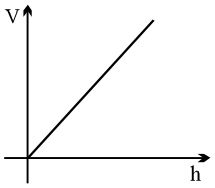
(C) $\vec{PC}$ (D) $\vec{PD}$

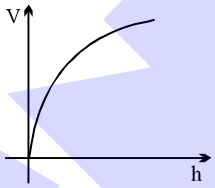
WA0031

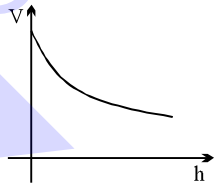
B. Speed of travelling wave

12. A uniform rope having some mass hanges vertically from a rigid support. A transverse wave pulse is produced at the lower end. The speed (v) of the wave pulse varies with height (h) from the lower end as:

(A) 

(B) 

(C) 

(D) 

WA0032

13. A wire is 4 m long and has a mass 0.2 kg. The wire is kept horizontally. A transverse pulse is generated by plucking one end of the taut (tight) wire. The pulse makes four trips back and forth along the cord in 0.8 sec. The tension is the cord will be :-

(A) 80 N (B) 160 N (C) 240 N (D) 320 N

WA0092

14. A transverse periodic wave on a string with a linear mass density of 0.200 kg/m is described by the following equation

$y = 0.05 \sin(420t - 21.0 x)$

where x and y are in metres and t is in seconds.

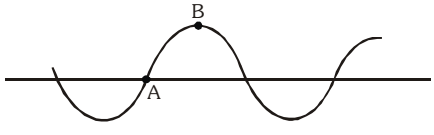
The tension in the string is equal to :

(A) 32 N (B) 42 N (C) 66 N (D) 80 N

WA0093

C. Energy & power in a travelling wave

15. A progressive wave is travelling in a string as shown. Then which of the following statement about KE and potential energy of the elements A and B is true?



(A) For point A : kinetic energy is maximum and potential energy is min.

(B) For point B : kinetic energy is minimum and potential energy is min.

(C) For point A : kinetic energy is minimum and potential energy is max.

(D) For point B : kinetic energy is minimum and potential energy is max.

WA0033

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16. Two waves are represented by the following equations

$y_1 = 5 \sin 2\pi\,(10t - 0.1x)$

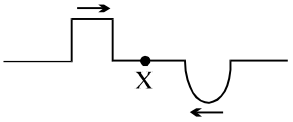
$y_2 = 10 \sin 2\pi\,(20t - 0.2x)$ Ratio of intensities I_2 / I_1 will be

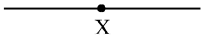
(A) 1 (B) 2 (C) 4 (D) 16

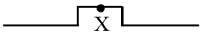
WA0094

D. Superposition of waves, reflection and transmission of waves

17. The diagram below shows two pulses traveling towards each other in a uniform medium with same speed. Pulses in the figure are at the same distance from X and has same height & width. Which diagram best represents the medium when the pulses meet at point X ?



(A) 

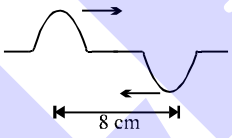
(B) 

(C) 

(D) 

WA0035

18. Two pulses in a stretched string whose centres are initially 8 cm apart are moving towards each other as shown in figure. The speed of each pulse is 2 cm/s. After 2 seconds, the total energy of the pulses will be :-

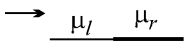


(A) zero (B) purely kinetic

(C) purely potential (D) partly kinetic and partly potential

WA0037

19. A string consists of two parts attached at x = 0. The right part of the string (x > 0) has mass μ_r per unit length and the left part of the string (x < 0) has mass μ_l per unit length. The string tension is T. If a wave of unit amplitude travels along the left part of the string, as shown in the figure, what is the amplitude of the wave that is transmitted to the right part of the string ?



(A) 1 (B) $\dfrac{2}{1+\sqrt{\mu_l / \mu_r}}$ (C) $\dfrac{2\sqrt{\mu_l / \mu_r}}{1+\sqrt{\mu_l / \mu_r}}$ (D) $\dfrac{\sqrt{\mu_l / \mu_r}-1}{\sqrt{\mu_l / \mu_r}+1}$

WA0039

20. A wave travels on a light string .The equation of the wave is Y= Asin(kx – wt + 30°). It is reflected from a heavy string tied to an end of the light string at x = 0.If 64% of the incident energy is reflected the equation of the reflected wave is

(A) Y = 0.8Asin (kx – wt + 30^0 + 180^0) (B) Y = 0.8Asin (kx + wt + 30^0 + 180^0)

(C) Y = 0.8Asin (kx + wt – 30^0) (D) Y = 0.8Asin (kx + wt + 30^0)

WA0040

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E. Standing waves

21. A wave is represented by the equation y = 10 sin 2π(100t – 0.02x) + 10 sin 2π(100t + 0.02x). The maximum amplitude and loop length are respectively

(A) 20 units and 30 units

(B) 20 units and 25 units

(C) 30 units and 20 units

(D) 25 units and 20 units

WA0041

22. Five waveforms moving with equal speeds on the x-axis

$y_1 = 8 \sin(\omega t + kx)$; $y_2 = 6 \sin(\omega t + \frac{\pi}{2} + kx)$; $y_3 = 4 \sin(\omega t + \pi + kx)$; $y_4 = 2 \sin(\omega t + \frac{3\pi}{2} + kx)$;

$y_5 = 4\sqrt{2} \sin(\omega t - kx + \frac{\pi}{4})$ are superimposed on each other. The resulting wave is :

(A) $8\sqrt{2} \cos kx \sin(\omega t + \frac{\pi}{4})$ (B) $8\sqrt{2} \sin(\omega t - kx + \frac{\pi}{4})$

(C) $8\sqrt{2} \sin kx \cos(\omega t + \frac{\pi}{4})$ (D) $8 \sin(\omega t + kx)$

WA0043

23. A wave represented by the equation y = A cos (kx – ωt) is superimposed with another wave to form a statioary wave such that the point x = 0 is a node. The equation of the other wave is:

(A) –A sin (kx + ωt) (B) – A cos (kx + ωt)

(C) A sin (kx + ωt) (D) A cos (kx + ωt)

WA0042

24. The equation of a standing wave produced on a string fixed at both the ends is -

$y = (0.4) \sin[(0.314\ cm^{-1})x] \cos[(600\ \pi\ s^{-1})t]$

What would be the smallest length of the string?

(A) 20 cm (B) 30 cm (C) 10 cm (D) 5 cm

WA0095

25. For the stationary wave $y = 4\sin(\pi x/15)\cos(96\pi t)$ the distance between a node and the next antinode is

(A) 7.5 (B) 15 (C) 22.5 (D) 30

WA0096

26. The vibrations of a string of length 60 cm fixed at both ends are represented by the equation $y = 4\sin(\pi x/15)\cos(96\pi t)$, where x and y are in cm and t in seconds. The maximum displacement at $x = 5$ cm is

(A) $2\sqrt{3}\,cm$ (B) $3\sqrt{2}\,cm$ (C) $\sqrt{2}\,cm$ (D) $\sqrt{3}\,cm$

WA0097

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27. A standing wave pattern of amplitude A in a string of length L shows 2 nodes (plus those at two ends). If one end of the string corresponds to the origin and v is the speed of progressive wave, the disturbance in the string, could be represented (with appropriate phase) as:

(A) $y(x,t) = A\sin\left(\frac{2\pi x}{L}\right)\cos\left(\frac{2\pi vt}{L}\right)$ (B) $y(x,t) = A\cos\left(\frac{3\pi x}{L}\right)\sin\left(\frac{3\pi vt}{L}\right)$

(C) $y(x,t) = A\cos\left(\frac{4\pi x}{L}\right)\cos\left(\frac{4\pi vt}{L}\right)$ (D) $y(x,t) = A\sin\left(\frac{3\pi x}{L}\right)\cos\left(\frac{3\pi vt}{L}\right)$

WA0098

28. A string clamped at both ends is vibrating. At the moment the string looks flat, the instantaneous transverse velocity of points along the string, excluding its end-points, must be

(A) zero everywhere

(B) dependent on the location along the string

(C) non zero everywhere

(D) non-zero and in the same direction everywhere

WA0047

29. A standing wave having 3 nodes and 2 antinodes is formed between two atoms having a distance of 1.21Å between them. The wavelength of the standing wave is

(A) 1.21Å (B) 2.42Å (C) 3.63Å (D) 6.05Å

WA0099

30. A vibrating stretched string resonates with a tuning fork of frequency 512 Hz when the length of the string is 0.5 m. The length of the string required to vibrate resonantly with a tuning fork of frequency 256 Hz would be

(A) 0.25 m (B) 0.75 m (C) 1.0 m (D) 2.0 m

WA0100

31. The fundamental frequency of a sonometer wire is n. If its length, diameter and tension are doubled, the material of the wire remaining the same, the new fundamental frequency will be

(A) n (B) $n/\sqrt{2}$ (C) $n/2$ (D) $\frac{n}{2\sqrt{2}}$

WA0101

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EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

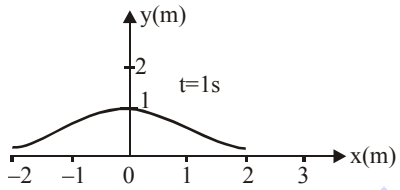
A. Equation of travelling wave

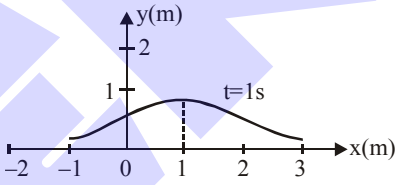
1. A siren can be made from a rotating flat disc, which has regularly speed holes punched through it along a circle concentric with the axis of rotation. An air nozzle is directed against the disc. Each time a hole passes the nozzle, a puff of air is released to generate a wave pulse. Sound of what frequency will be produced by disc containing 72 holes and rotating at 1800 rev.min.?

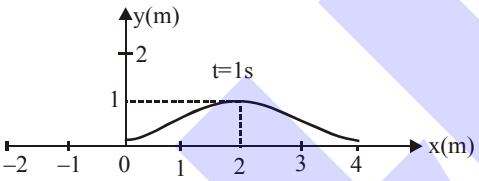
(A) $72\times1800\,Hz$ (B) $72\times1800\times60\,Hz$ (C) 2160 Hz (D) 72 Hz

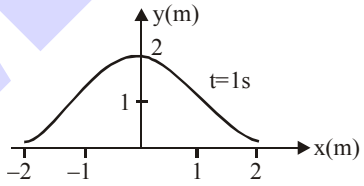
WA0102

2. A wave pulse is given by the equation $y = f(x, t) = A \exp(-B(x-vt)^2)$. Given A =1.0m, B = 1.0 m^{-2} and v = + 2.0 m/s, which of the following graph shows the correct wave profile at the instant t = 1 s?

(A) 

(B) 

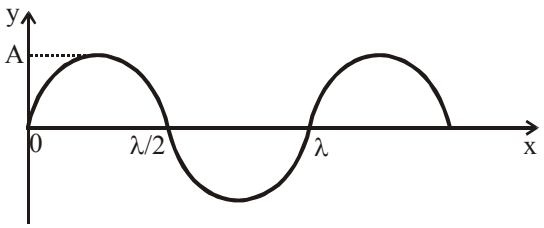
(C) 

(D) 

WA0027

3. Here given snap shot of a progressive wave at t = 0 with time period = T. Then the equation of the wave if wave is going in +ve x-direction and if wave is going in –ve x-direction will be respectively.

$\left(\text{Here, } T = \frac{2\pi}{\omega}\right)$



(A) $y = A\sin(kx + \omega t)$, $y = A\sin(kx - \omega t)$ (B) $y = A\cos(kx + \omega t)$, $y = A\cos(kx - \omega t)$

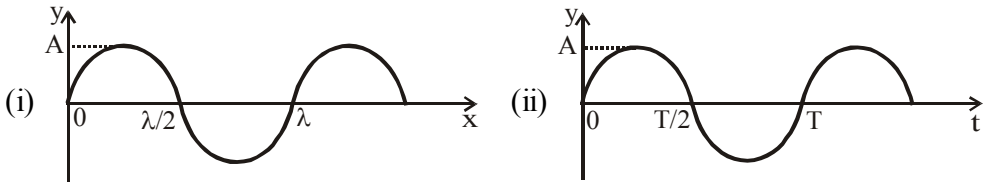
(C) $y = A\sin(\omega t - kx)$, $y = A\sin(\omega t + kx)$ (D) $y = A\sin(kx - \omega t)$, $y = A\sin(kx + \omega t)$

WA0029

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4. Here given figure (i) shows snap shot at t = T/4 and figure (ii) shows motion of particle at x = λ/4.

Then the possible equations of the wave will be $\left(\text{Here, } T = \frac{2\pi}{\omega}\right)$:-



(A) $y = A \sin\left(\omega t + kx - \frac{\pi}{2}\right)$

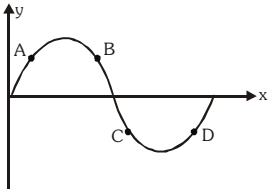
(B) $y = A \sin\left(\omega t - kx + \frac{\pi}{2}\right)$

(C) Both $y = A \cos\left(\omega t + kx - \frac{\pi}{2}\right)$ and $y = A \cos\left(\omega t - kx + \frac{\pi}{2}\right)$

(D) Both $y = A \sin\left(\omega t + kx - \frac{\pi}{2}\right)$ and $y = A \sin\left(\omega t - kx + \frac{\pi}{2}\right)$

WA0044

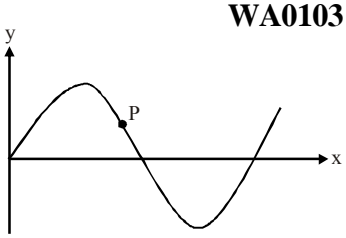
5. A wave is moving towards positive x-axis as shown in figure. Then the point (s) at which acceleration and velocity of particle are parallel to each other :-



(A) A,D
(B) B,C
(C) A
(D) A,C

WA0103

6. A transverse sinusoidal wave moves along a string in the positive x-direction at a speed of 10 cm/s. The wavelength of the wave is 0.5 m and its amplitude is 10 cm. At a particular time t, the snap-shot of the wave is shown in figure. The velocity of point P when its displacement is 5 cm is : **[IIT-JEE 2008]**



(A) $\frac{\sqrt{3}\pi}{50}\hat{j}$ m/s (B) $-\frac{\sqrt{3}\pi}{50}\hat{j}$ m/s (C) $\frac{\sqrt{3}\pi}{50}\hat{i}$ m/s (D) $-\frac{\sqrt{3}\pi}{50}\hat{i}$ m/s

WA0104

7. The transverse displacement y(x, t) of a wave on a string is given by $y(x,t) = e^{-\left(ax^2 + bt^2 + 2\sqrt{ab}xt\right)}$. This represents a :- **[AIEEE - 2011]**

(A) standing wave of frequency $\sqrt{b}$

(B) standing wave of frequency $\frac{1}{\sqrt{b}}$

(C) wave moving in +x directionwith speed $\sqrt{\frac{a}{b}}$

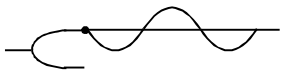
(D) wave moving in –x direction with speed $\sqrt{\frac{b}{a}}$

WA0060

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound\Wave-EM\Wave & Wave Op\Eng\01_WOS\01_Th+Ex.p65

C. Energy & power in a travelling wave

8. The prong of a electrically operated tuning fork is connected to a long string of $\mu = 1$ kg/m and tension 25N. The maximum velocity of the prong is 1 cm/s, then the average power needed to drive the prong is:

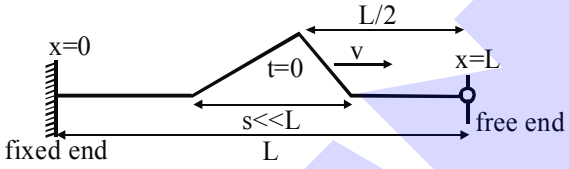


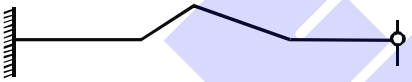
(A) 5×10^{-4} W (B) 2.5×10^{-4} W (C) 1×10^{-4} W (D) 10^{-3} W

WA0034

D. Superposition of waves, reflection and transmission of waves

9. A small pulse travelling with speed v in a string is shown at t = 0, moving towards free end. Which of these is not **CORRECTLY** matched.



(i) $t = \frac{L}{v}$ (P) 

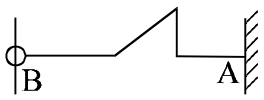
(ii) $t = \frac{2L}{v}$ (Q) 

(iii) $t = \frac{3L}{v}$ (R) 

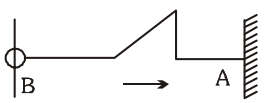
(A) (i) (B) (ii) (C) (iii) (D) None of these

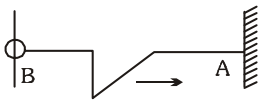
WA0038

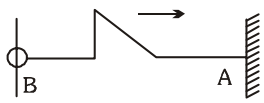
10. A pulse shown here is reflected from the rigid wall A and then from free end B. The shape of the string after these 2 reflection will be



(A) 

(B) 

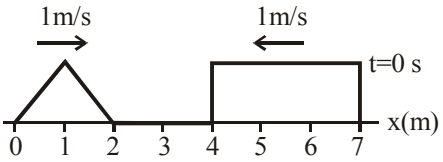
(C) 

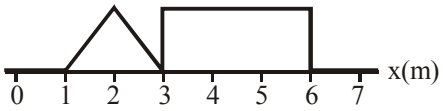
(D) 

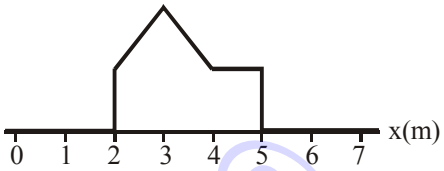
WA0105

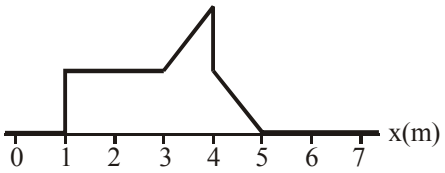
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound\Wave-EM Wave & Wave Op\Eng\01_WOS\01_Th + Ex.p65

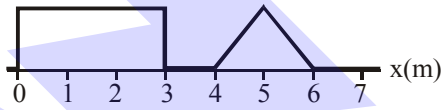
11. Two pulses on a string approach each other at speeds of 1 m/s. What is the shape of the string at t = 3s:-



(A) 

(B) 

(C) 

(D) 

WA0106

12. **Statement-1 :** When a pulse on string reflects from free end, the resultant pulse is formed in such a way that slope of string at free end is zero.

Statement-2 : Zero resultant slope ensures that there is no force component perpendicular to string.

(A) Statement-1 is True, Statement-2 is True, Statement-2 is a correct explanation for statement-1

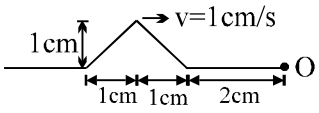
(B) Statement-1 is True, Statement-2 is True, Statement-2 is NOT a correct explanation for Statement-1

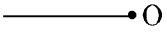
(C) Statement-1 is True, Statement-2 is False

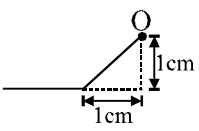
(D) Statement-1 is False, Statement-2 is True

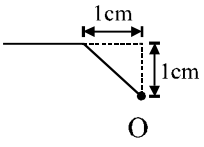
WA0107

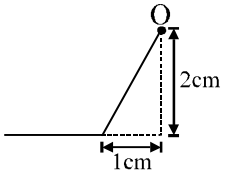
13. A wave pulse on a string has the dimension shown in figure. The waves speed is v = 1 cm/s. If point O is a free end. The shape of wave at time t = 3 s is :



(A) 

(B) 

(C) 

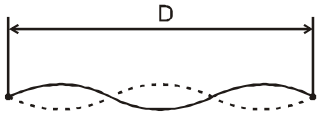
(D) 

WA0108

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound\Wave-BM\Wave & Wave Op\Eng\01_WOS\01_Th + Ex.p65

E. Standing waves

14. A nylon guitar string has a linear density of 7.20 g/m and is under a tension of 150 N. The fixed supports are distance D = 90.0 cm apart. The string is oscillating in the standing wave pattern shown in figure. Calculate the



(i) The speed of the traveling waves whose superposition gives this standing wave

(A) $\frac{125}{\sqrt{3}}$ m/s (B) $\frac{500}{\sqrt{3}}$ m/s (C) $\frac{250}{\sqrt{3}}$ m/s (D) $100\sqrt{3}$ m/s

(ii) The wavelength of the traveling waves whose superposition gives this standing wave.

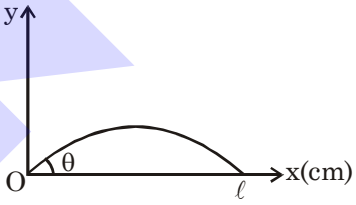
(A) 20 cm (B) 40.0 cm (C) 60.0 cm (D) 80.0 cm

(iii) The frequency of the traveling waves whose superposition gives this standing wave.

(A) $\frac{1000}{3\sqrt{3}}$ Hz (B) $\frac{1250}{3\sqrt{3}}$ Hz (C) $\frac{1500}{3\sqrt{3}}$ Hz (D) $\frac{1750}{3\sqrt{3}}$ Hz

WA0109

15. A guitar string is 80 cm long and has a fundamental frequency of 400 Hz. In its fundamental mode the maximum displacement is 2mm at the middle. If the tension in the string is 10N, what is the maximum value of that component of the force on the end support which is perpendicular to the equilibrium position of the string ?



(A) $\frac{\pi}{40}$N (B) $\frac{\pi}{80}$N (C) $\frac{\pi}{20}$N (D) $\frac{\pi}{10}$N

WA0110

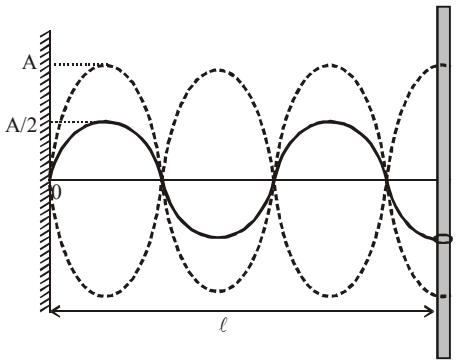
16. Here given snap shot at $t = \frac{T}{12}$ of a standing wave. Then the equations of the wave will be when particles are moving towards their extreme and when particles are moving towards the mean position respectively $\left(\text{Here, } T = \frac{2\pi}{\omega}\right)$.

(A) $y = A \sin kx \sin \omega t$, $y = A \sin\left(\omega t + \frac{2\pi}{3}\right) \sin kx$

(B) $y = A \sin kx \cos \omega t$, $y = A \sin\left(\omega t + \frac{2\pi}{3}\right) \sin kx$

(C) $y = A \cos kx \cos \omega t$, $y = A \cos\left(\omega t + \frac{2\pi}{3}\right) \sin kx$

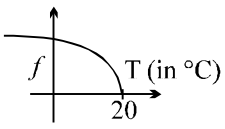
(D) $y = A \cos kx \sin \omega t$, $y = A \cos\left(\omega t + \frac{2\pi}{3}\right) \cos kx$

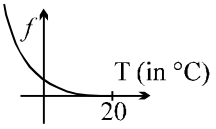


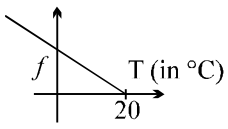
WA0111

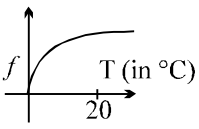
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound\Wave-EM Wave & Wave Op\Eng\01_WOS\01_Th+Ex.p65

17. A metal wire is clamped between two vertical walls. At 20 °C the unstrained length of the wire is exactly equal to the separation between walls. If the temperature of the wire is decreased the graph between fundamental frequency (f) and temperature (T) of the wire is

(A) 

(B) 

(C) 

(D) 

WA0045

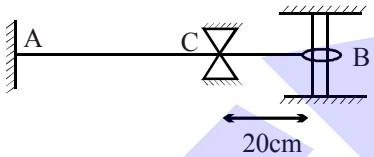
18. What is the fractional change in the tension necessary in a sonometer of fixed length to produce a note one octave lower (half of original frequency) than before

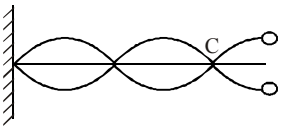
(A) 1/4 (B) 1/2 (C) 2/3 (D) 3/4

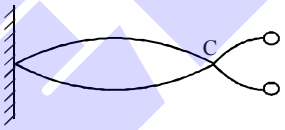
(E) 2/1

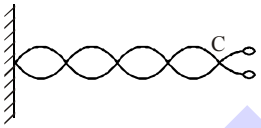
WA0046

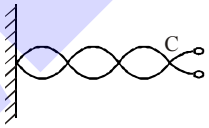
19. A 1m long wire having tension T is fixed at A and free at B. The point C, 20 cm from B is constrained to be stationary. What is shape of string for fundamental mode?



(A) 

(B) 

(C) 

(D) 

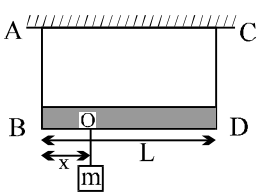
WA0048

20. A sonometer wire has a total length of 1 m between the fixed ends. Two wooden bridges are placed below the wire at a distance 1/7 m from one end and 4/7 m from the other end. The three segments of the wire have their fundamental frequencies in the ratio :

(A) 1 : 2 : 3 (B) 4 : 2 : 1 (C) 1 : 1/2 : 1/3 (D) 1 : 1 : 1

WA0112

21. A massless rod BD is suspended by two identical strings AB and CD of equal length. A block of mass m is suspended from point O such that BO is equal to 'x'. Further, it is observed that the frequency of 1st harmonic (fundamental frequency) in AB is equal to 2nd harmonic frequency in CD. Then, length of BO is **[IIT-JEE 2006]**



(A) $\frac{L}{5}$ (B) $\frac{L}{4}$ (C) $\frac{4L}{5}$ (D) $\frac{3L}{4}$

WA0113

22. The ends of a stretched wire of length L are fixed at x = 0 and x = L. In one experiment, the displacement of the wire is $y_1 = A \sin(\pi x/L) \sin \omega t$ and energy is E_1 and in another experiment its displacement is $y_2 = A\sin(2\pi x/L) \sin 2\omega t$ and energy is E_2. Then **[JEE 2001 (Scr)]**

(A) $E_2 = E_1$ (B) $E_2 = 2E_1$ (C) $E_2 = 4E_1$ (D) $E_2 = 16E_1$

WA0049

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound\Wave-EM\Wave & Wave Op\Eng\01_WOS\01_Th + Ex.p65

23. A travelling wave represented by $y = A\sin(\omega t - kx)$ is superimposed on another wave represented by $y = A\sin(\omega t + kx)$. The resultant is :- **[AIEEE-2011]**

(A) A standing wave having nodes at $x = \left(n + \frac{1}{2}\right)\frac{\lambda}{2}, n = 0,1,2$

(B) A wave travelling along + x direction

(C) A wave travelling along –x direction

(D) A standing wave having nodes at $x = \frac{n\lambda}{2}$; n = 0, 1, 2

WA0061

MULTIPLE CORRECT TYPE QUESTIONS

A. Equation of travelling wave

24. $Y(x,t) = 0.05 / [(4x + 2t)^2 + 5]$ represents a moving wave pulse, where x and y are in meters and t is in seconds. Then which statement(s) are **CORRECT**:

(A) Pulse is moving in – x direction (B) Wave speed is 0.5 m/s

(C) Maximum particle displacement is 1 cm (D) It is a symmetric pulse

WA0114

25. Consider a hypothetical wave pulse (at time t = 0) given by the following (y, x in meter)

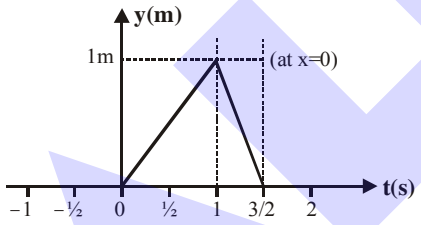
$y = 0, x < 0$

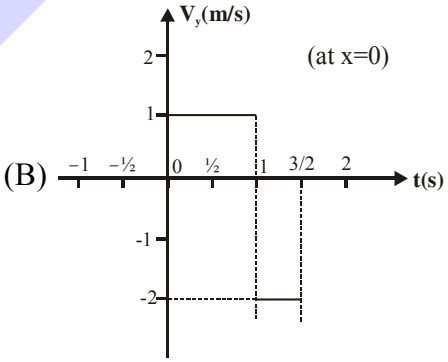
$y = x/2, 2 > x \geq 0$

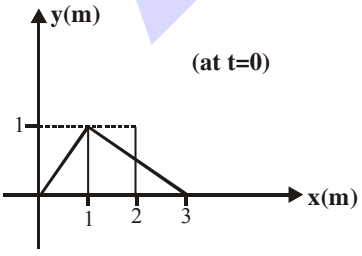
$y = 3 - x, 3 \geq x \geq 2$

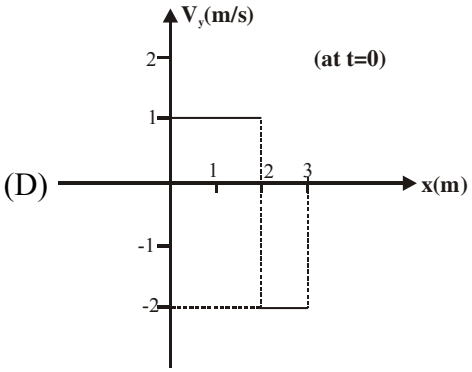
$y = 0, x > 3$

The pulse travels leftwards (negative x direction) at speed v = 2 m/s. Which of the following plots are correct? [V_y is the velocity of particle]

(A) 

(B) 

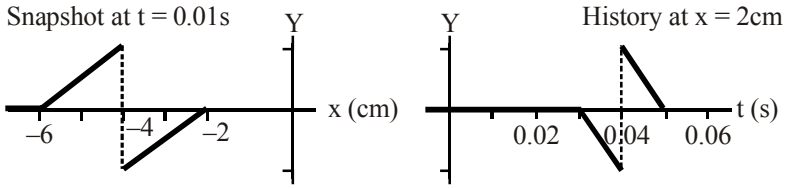
(C) 

(D) 

WA0050

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound\Wave-EM\Wave & Wave Op\Eng\01_WOS\01_Th+Ex.p65

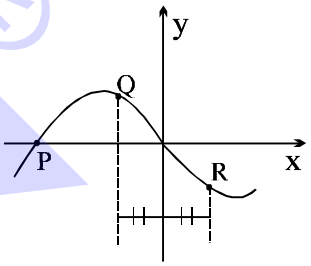
26. Figure shows a snapshot graph and a history graph for a wave pulse on a stretched string. They describe the same wave from two perspectives.



(A) the wave is travelling in positive x-direction
(B) the wave is travelling in negative x-direction
(C) the speed of the wave is 2 m/s.
(D) the peak is located at x = –6 cm at t = 0.

WA0051

27. At a certain moment, the photograph of a string on which a harmonic wave is travelling to the right is shown. Then, which of the following is true regarding the velocities of the points P, Q and R on the string.

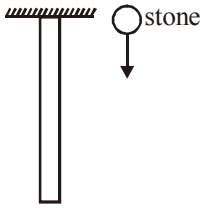


(A) v_P is upwards
(B) $v_Q = -v_R$
(C) $|v_P| > |v_Q| = |v_R|$
(D) $v_Q = v_R$

WA0052

B. Speed of travelling wave

28. A uniform wire of length ℓ and mass m is hanging from a fixed support as shown. A transverse wave pulse is produced at the top of rod, and a stone is dropped simultaneously from top. Then :-



(A) Stone will definitely overtake the pulse, irrespective of length of wire.
(B) Location of overtake will be independent of mass of wire, provided length remains same.
(C) As the wavepulse travels downward its length of pulse decreases.
(D) As the wavepulse travels downward its length of pulse increases.

WA0115

C. Energy & power in a travelling wave

29. In a travelling one dimensional mechanical sinusoidal, wave
(A) potential energy and kinetic energy of an element become maximum simultaneously.
(B) all particles oscillate with the same frequency and the same amplitude
(C) all particles may come to rest simultaneously
(D) we can find two particles, in a length equal to half of a wavelength, which have the same non zero acceleration simultaneously.

WA0054

E. Standing waves

30. An string has resonant frequencies given by 1001 Hz and 2639 Hz.
(A) If the string is fixed at one end only, 910 Hz can be a resonance frequency.
(B) If the string is fixed at one end only, 1911 Hz can be a resonance frequency.
(C) If the string is fixed at both the ends, 364 Hz can be one of the resonant frequency.
(D) 1001 Hz is definitely not the fundamental frequency of the string.

WA0053

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound\Wave-EM\Wave & Wave Op\Eng\01_WOS\01_Th + Ex.p65

31. If the tension in a stretched string fixed at both ends is changed by 20%, the fundamental frequency is found to increase by 15Hz. then the
(A) original frequency is 157 Hz
(B) velocity of propagation of the transverse wave along the string changes by 5%
(C) velocity of propagation of the transverse wave along the string changes by 10%.
(D) fundamental wave length on the string does not change.

WA0116

32. A standing wave of time period T is set up in a string clamped between two rigid supports. At t = 0 antinode is at its maximum displacement 2A.
(A) The energy density of a node is equal to energy density of an antinode for the first time at t = T/4.
(B) The energy density of node and antinode becomes equal after T/2 second.
(C) The displacement of the particle at antinode at $t = \frac{T}{8}$ is $\sqrt{2}A$
(D) The displacement of the particle at node is zero

WA0117

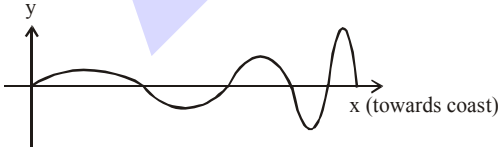
COMPREHENSION TYPE QUESTIONS

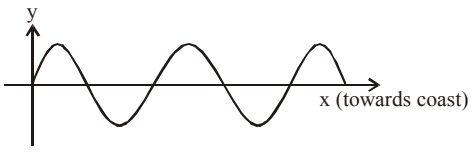
A. Equation of travelling wave

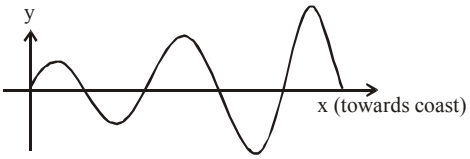
Paragraph for Question no. 33 to 35

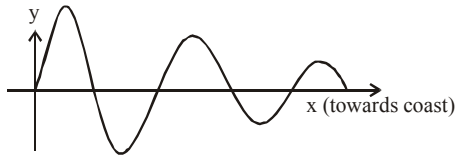
The phenomenon we call tsunami is a series of large waves of extremely long wavelength and period usually generated by a violent, impulsive undersea disturbance or activity near the coast or in the ocean. Once a tsunami has been generated, its energy is distributed throughout the water column, regardless of the ocean's depth. The waves will travel outward on the surface of the ocean in all direction away from the source area, much like the ripples caused by throwing a rock into a pond. The deeper the water, the greater the speed of tsunami waves will be. For example, at the deepest ocean depths the tsunami wave speed will be as much as 800 km/hr.
Tsunamis arrive at a coastline as a series of successive crests (high water levels) and troughs (low water levels) – usually occurring 10 minutes apart. As they enter the shallow waters of coastlines, bays, or harbors, their speed decreases to 50 km/hr. However 100 or more kilometers away, another tsunami wave travels in deep water towards the same shore at a much greater speed, and still behind it there is another wave, traveling at even greater speed. As the tsunami waves become compressed near the coast, the wavelength is shortened and the wave energy is directed upward-thus increasing their heights considerably. Just as with ordinary surf, the energy of the tsunami waves must be contained in a smaller volume of water, so the waves grow in height.

33. The graph of *Tsunami* wave as it approaches the coast is

(A) 

(B) 

(C) 

(D) 

WA0118

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34. The ratio of wavelength of *Tsunami* wave at centre and at coast is

(A) 20 (B) 16 (C) 5 (D) 12

WA0119

35. Frequency at centre of *Tsunami* will be

(A) 8.33×10^{-4} Hz (B) 12.33×10^{-4} Hz

(C) 8.33×10^{4} Hz (D) 12.33×10^{4} Hz

WA0120

MATRIX MATCH TYPE QUESTIONS

E. Standing waves

36. In a string a standing wave is set up whose equation is given as $y = 2A \sin kx \cos \omega t$. The mass per unit length of the string is μ.

	Column-I		**Column-II**
(A)	at $t = 0$	(P)	Total energy per unit length at $x = 0$ is $2\mu A^2\omega^2$.
(B)	at $t = \frac{T}{8}$	(Q)	Total energy per unit length at $x = \lambda/4$ is $2\mu A^2\omega^2$.
(C)	at $t = \frac{T}{4}$	(R)	Total energy per unit length at $x = \lambda$ is $2\mu A^2\omega^2$.
(D)	at $t = \frac{T}{2}$	(S)	power transmitted through a point at $x = \lambda$ is 0.
		(T)	power transmitted through a point at $x = \lambda/4$ is 0.

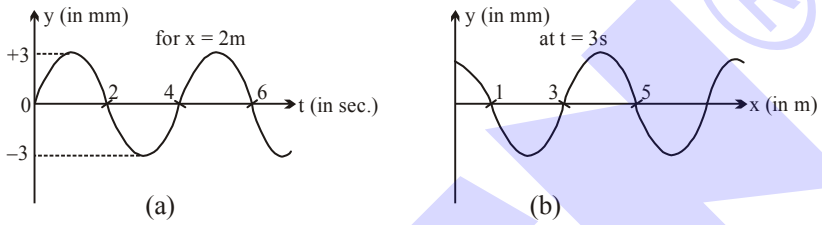
WA0058

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound\Wave-EM\Wave & Wave Op\Eng\01_WOS\01_Th + Ex.p65

EXERCISE (S)

A. Equation of travelling wave

1. A sinusoidal wave propagates along a string. In figure (a) and (b) 'y' represents displacement of particle from the mean position. 'x' & 't' have usual meanings. Find :
(a) wavelength, frequency and speed of the wave.
(b) maximum velocity and maximum acceleration of the particles
(c) the magnitude of slope of the string for x = 2 m at t = 4 sec.

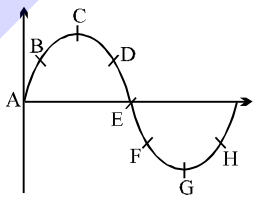


WA0002

2. A plane progressive wave of frequency 50 Hz, travelling along positive x-axis is represented as $y = (5 \times 10^{-5}\ m) \sin (100\ \pi t)$ at x = 0, wave speed is 300 m/s. Maximum difference in displacements at x = 0 and x = –3 m at any instant is Δy (in mm). Write $10\Delta y$ in OMR sheet.

WA0121

3. A transverse wave is travelling along a string from left to right. The fig. represents the shape of the string (snap-shot) at a given instant. At this instant (a) which points have an upward velocity (b) which points will have downward velocity (c) which points have zero velocity (d) which points have maximum magnitude of velocity.



WA0001

4. A light pointer fixed to one prong of a tuning fork touches a vertical plate. The fork is set vibrating and plate is allowed to fall freely. 8 complete oscillations are counted when the plate falls through 10 cm. What is the frequency of the tuning fork ?

WA0003

B. Speed of travelling wave

5. The extension in a string, obeying Hooke's law is x. The speed of wave in the stretched string is v. If the extension in the string is increased to 1.5 x find the new speed of wave.

WA0006

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6. The speed of transverse wave, going on a wire having a length 50 cm and mass 5.0 kg, is 80 m/s. The area of cross-section of the wire is 1.0 mm^2 and its young's modulus is 16×10^{11} N/m^2. Find the extension of the wire over its natural length.

WA0122

7. A copper wire is held at the two ends by rigid supports. At 30°C, the wire is just taut, with negligible. Find the speed of transverse waves in this wire at 10°C.
Given : Young modulus of copper = 1.3×10^{11} N/m^2.
Coefficient of linear expansion of copper = 1.7×10^{-5} $°C^{-1}$.
Density of copper = 9×10^3 kg/m^3. **[IIT-1979]**

WA0008

8. A uniform rope of mass m and length L hangs from a ceilling. (a) Show that the speed of a transverse wave on the rope is a funciton of y, the distance from the lower end, and is given by $v = \sqrt{gy}$. (b) Show that the time a transverse wave takes to travel the length of the rope is given by $t = 2\sqrt{L/g}$.

WA0123

9. A non-uniform rope of mass M and length L has a variable linear mass density given by $\mu = kx$ where x is the distance from one end of the wire and k is a constant.
(a) Show that $M = kL^2/2$
(b) Show that the time required for a pulse generated at one end of the wire to travel to the other end is given by $t = \sqrt{8ML/9F}$ where F (constant)is the tension in the wire.

WA0022

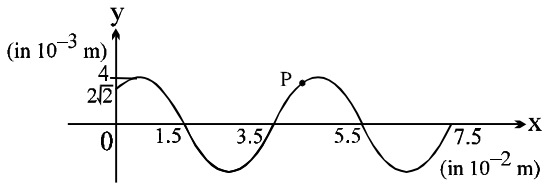
C. Energy & power in a travelling wave

10. A long string under tension of 100 N has one end at x = 0. A sinusoidal wave is generated at x = 0 whose equation is given by $y = (0.01 \text{ cm}) \sin\left[\left(\frac{\pi x}{10} \text{ m}\right) - 50\pi t \text{ (sec)}\right]$

(i) Sketch the shape of the string at $t = \frac{1}{50}$ sec.
(ii) Find the average power transmitted by the wave.
(iii) Draw velocity time graph of particle at x = 5 m.

WA0014

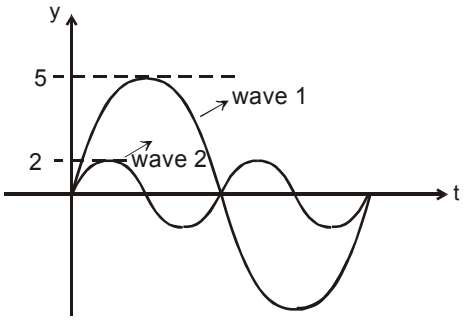
11. The figure shows a snap photograph of a vibrating string at t = 0. The particle P is observed moving up with velocity 20π cm/s. The angle made by string with x-axis at P is 6°.
(a) Find the direction in which the wave is moving
(b) the equation of the wave
(c) the total energy carried by the wave per cycle of the string, assuming that μ, the mass per unit length of the string = 50 gm/m.



WA0005

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12. Two waves in the same medium are represented by y-t curves in the figure. Find ratio of their average intensities?



WA0124

13. A transverse wave of amplitude 0.50 mm and frequency 100 Hz is produced on a wire stretched to a tension of 100 N. If the wave speed is 100 m/s, what average power is the source transmitting to the wire?

WA0125

D. Superposition of waves, reflection and transmission of waves

14. Two waves passing through a region are represented by

$y = (1.0\text{ cm}) \sin [(3.14\text{ cm}^{-1})\, x - (157\text{ s}^{-1})\, t]$

and $y = (1.5\text{ cm}) \sin [(1.57\text{ cm}^{-1})x - (314\text{ s}^{-1})\, t]$.

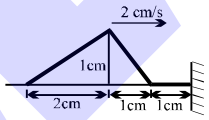
Find the displacement of the particle at x = 4.5 cm at time t = 5.0 ms.

WA0126

15. The figure shown a triangular pulse on a rope at t = 0. It is approaching a fixed end at 2 cm/s

(a) Draw the pulse at t = 2 sec.

(b) The particle speed on the leading edge at the instant depicted is _____.



WA0010

16. A plane wave given by equation $y = 0.04 \sin (0.5\pi x - 100\pi t)$, where x and y are in meter and t in sec is incident normally on a boundary between two media beyond which wave speed becomes doubled. State boundary condition and find the equation of the reflected and transmitted waves. Take x = 0 as the boundary between two media.

WA0012

E. Standing waves

17. A 20 cm long string having a mass of 1.0 g, is fixed at both the ends. The tension in the string is 0.5N. The string is set into vibrations using an external vibrator of frequency 100 Hz. Find the separation (in cm) between the successive nodes on the string. **[IIT-JEE 2009]**

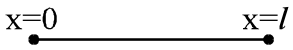
WA0127

18. A 40 cm long wire having a mass 3.2 gm and area of cross section 1 mm^2 is stretched between the supports 40.05 cm apart. In its fundamental mode, it vibrates with a frequency 1000/64 Hz. Find the young's modulus of the wire.

WA0128

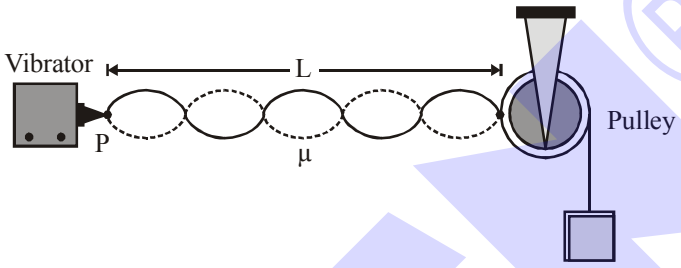
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound\Wave-EM Wave & Wave Op\Eng\01_WOS\01_Th+Ex.p65

19. A string between x = 0 and x = l vibrates in fundamental mode. The amplitude A, tension T and mass per unit length μ is given. Find the total energy of the string. **[IIT-JEE 2003(Scr)]**



WA0013

20. In the arrangement shown in figure, a mass can be hung from a string (with a linear mass density of μ = 0.00100 kg/m) that passes over a light pulley. The string is connected to a vibrator (of constant frequency f), and the length of the string between point P and the pulley is L = 2.00 m. When the mass m is either 16.0 kg or 25.0 kg, standing wave are observed; however, no standing waves are observed with any mass between these values. What is the frequency of the vibrator (in Hz) ?

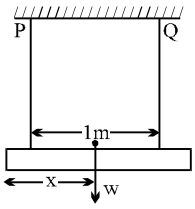


WA0129

21. The length of the sonometer wire between two fixed ends is 100 cm. Where should the two bridges be placed so as to divide the wire into three segments whose fundamental frequencies are in the ratio of 1 : 2 : 3 ?

WA0130

22. A 1m long rod of negligible weight is suspended by two wires P & Q of equal length as shown in figure. The cross-sectional areas are 1.5 cm^2 & 1 cm^2. The densities of these materials are 7.8 gm/cc & 2.6 gm/cc respectively. At what point along the rod should a weight w be suspended in order to produce equal stress in the strings. What is the ratio of velocities of transverse waves in these strings.



WA0131

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EXERCISE-JM

1. A sonometer wire of length 1.5m is made of steel. The tension in it produces an elastic strain of 1%. What is the fundamental frequency of steel if density and elasticity of steel are 7.7×10^3 kg/m^3 and 2.2×10^{11} N/m^2 respectively ? **[JEE-Main-2013]**

(1) 188.5 Hz (2) 178.2 Hz (3) 200.5 Hz (4) 770 Hz

WA0062

2. A uniform string of length 20m is suspended from a rigid support. A short wave pulse is introduced at its lowest end. It starts moving up the string. The time taken to reach the support is :-
(take g = 10 ms^{-2}) **[JEE-Main-2016]**

(1) $\sqrt{2}$ s (2) $2\pi\sqrt{2}$ s (3) 2s (4) $2\sqrt{2}$ s

WA0063

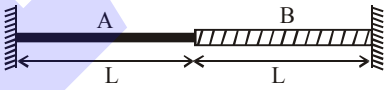
SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

3. A heavy ball of mass M is suspended from the ceiling of a car by a light string of mass m (m<<M). When the car is at rest, the speed of transverse waves in the string is 60 ms^{-1}. When the car has acceleration a, the wave-speed increases to 60.5 ms^{-1}. The value of a, in terms of gravitational acceleration g, is closest to : **[JEE-Main-2019_Jan]**

(1) $\frac{g}{5}$ (2) $\frac{g}{20}$ (3) $\frac{g}{10}$ (4) $\frac{g}{30}$

WA0080

4. A wire of length 2L, is made by joining two wires A and B of same length but different radii r and 2r and made of the same material. It is vibrating at a frequency such that the joint of the two wires forms a node. If the number of antinodes in wire A is p and that in B is q then the ratio p : q is : **[JEE-Main-2019_April]**



(1) 4 : 9 (2) 3 : 5 (3) 1 : 4 (4) 1 : 2

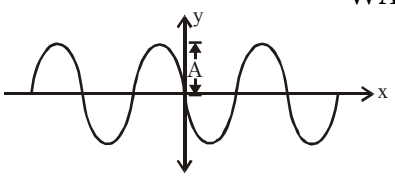
WA0081

5. A string is clamped at both the ends and it is vibrating in its 4th harmonic. The equation of the stationary wave is Y = 0.3 sin(0.157x) cos(200πt). The length of the string is : (All quantities are in SI units.) **[JEE-Main-2019_April]**

(1) 20 m (2) 80 m (3) 60 m (4) 40 m

WA0082

6. A progressive wave travelling along the positive x-direction is represented by y(x, t) = A sin (kx – ωt + ϕ). Its snapshot at t = 0 is given in the figure:
For this wave, the phase ϕ is : **[JEE-Main-2019_April]**



(1) 0 (2) $-\frac{\pi}{2}$ (3) π (4) $\frac{\pi}{2}$

WA0083

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7. A transverse wave travels on a taut steel wire with a velocity of v when tension in it is 2.06×10^4 N. When the tension is changed to T, the velocity changed to v/2. The value of T is close to : **[JEE-Main-2020_Jan]**

(1) 10.2×10^2 N (2) 5.15×10^3 N (3) 2.50×10^4 N (4) 30.5×10^4 N

WA0084

8. Which of the following equations represents a travelling wave ? **[JEE-Main-2021_Feb]**

(1) $y = A\sin(15x - 2t)$ (2) $y = Ae^{-x^2}(vt + \theta)$ (3) $y = Ae^x\cos(\omega t - \theta)$ (4) $y = A\sin x \cos\omega t$

WA0132

9. A sound wave of frequency 245 Hz travels with the speed of 300 ms^{-1} along the positive x-axis. Each point of the wave moves to and fro through a total distance of 6 cm. What will be the mathematical expression of this travelling wave ? **[JEE-Main-2021_March]**

(1) $Y(x,t) = 0.03\ [\sin 5.1\ x - (0.2 \times 10^3)t]$ (2) $Y(x,t) = 0.06\ [\sin 5.1\ x - (1.5 \times 10^3)t]$

(3) $Y(x,t) = 0.06\ [\sin 0.8\ x - (0.5 \times 10^3)t]$ (4) $Y(x,t) = 0.03\ [\sin 5.1\ x - (1.5 \times 10^3)t]$

WA0133

10. Two travelling waves produces a standing wave represented by equation,
$y = 1.0$ mm $\cos(1.57\ cm^{-1})\ x \sin(78.5\ s^{-1})t$.
The node closest to the origin in the region $x > 0$ will be at x = ... cm. **[JEE-Main-2021_Aug]**

WA0134

11. A wire having a linear mass density 9.0×10^{-4} kg/m is stretched between two rigid supports with a tension of 900 N. The wire resonates at a frequency of 500 Hz. The next higher frequency at which the same wire resonates is 550 Hz. The length of the wire is _____m. **[JEE-Main-2021_Aug]**

WA0135

12. A transverse wave is represented by $y = 2\sin(\omega t - kx)$ cm. The value of wavelength (in cm) for which the wave velocity becomes equal to the maximum particle velocity, will be; **[JEE-Main-2022_July]**

(A) 4π (B) 2π (C) π (D) 2

WA0136

13. A wire of length 30 cm, stretched between rigid supports, has it's n^{th} and $(n+1)^{th}$ harmonics at 400Hz and 450 Hz, respectively. If tension in the string is 2700 N, it's linear mass density is……..kg/m. **[JEE-Main-2022_July]**

WA0137

14. The speed of a transverse wave passing through a string of length 50 cm and mass 10 g is 60 ms^{-1}. The area of cross-section of the wire is 2.0 mm^2 and its Young's modulus is 1.2×10^{11} Nm^{-2}. The extension of the wire over its natural length due to its tension will be $x \times 10^{-5}$ m. The value of x is_____. **[JEE-Main-2022_July]**

WA0138

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EXERCISE (JA)

1. A horizontal stretched string, fixed at two ends, is vibrating in its fifth harmonic according to the equation, $y(x, t) = (0.01m) \sin[(62.8\ m^{-1})x]\cos[(628\ s^{-1})t]$. Assuming $\pi = 3.14$, the correct statement(s) is (are) **[JEE-Advance-2013]**

(A) The number of nodes is 5.

(B) The length of the string is 0.25 m.

(C) The maximum displacement of the midpoint of the string, from its equilibrium position is 0.01m.

(D) The fundamental frequency is 100 Hz.

WA0065

2. One end of a taut string of length 3m along the x–axis is fixed at x = 0. The speed of the waves in the string is 100 ms^{-1}. The other end of the string is vibrating in the y direction so that stationary waves are set up in the string. The possible waveform (s) of these stationary waves is(are) :- **[JEE-Advance-2014]**

(A) $y(t) = A\sin\frac{\pi x}{6}\cos\frac{50\pi t}{3}$

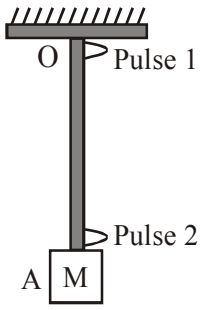
(B) $y(t) = A\sin\frac{\pi x}{3}\cos\frac{100\pi t}{3}$

(C) $y(t) = A\sin\frac{5\pi x}{6}\cos\frac{250\pi t}{3}$

(D) $y(t) = A\sin\frac{5\pi x}{2}\cos 250\pi t$

WA0066

3. A block M hangs vertically at the bottom end of a uniform rope of constant mass per unit length. The top end of the rope is attached to a fixed rigid support at O. A transverse wave pulse (Pulse 1) of wavelength λ_0 is produced at point O on the rope. The pulse takes time T_{OA} to reach point A. If the wave pulse of wavelength λ_0 is produced at point A (Pulse 2) without disturbing the position of M it takes time T_{AO} to reach point O. Which of the following options is/are **correct** ? **[JEE-Advance-2017]**



(A) The time $T_{AO} = T_{OA}$

(B) The velocities of the two pulses (Pulse 1 and Pulse 2) are the same at the midpoint of rope

(C) The wavelength of Pulse 1 becomes longer when it reaches point A

(D) The velocity of any pulse along the rope is independent of its frequency and wavelength.

WA0067

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4. Answer the following by appropriately matching the lists based on the information given in the paragraph.

A musical instrument is made using four different metal strings, 1,2,3 and 4 with mass per unit length μ, 2μ, 3μ and 4μ respectively. The instrument is played by vibrating the strings by varying the free length in between the range L_0 and $2L_0$. It is found that in string-1 (μ) at free length L_0 and tension T_0 the fundamental mode frequency is f_0.

List-I gives the above four strings while list-II lists the magnitude of some quantity.

If the tension in each string is T_0, the correct match for the highest fundamental frequency in f_0 units will be, **[JEE-Advance-2019]**

	List-I		**List-II**
(I)	String-1(μ)	(P)	1
(II)	String-2 (2μ)	(Q)	1/2
(III)	String-3 (3μ)	(R)	$1/\sqrt{2}$
(IV)	String-4 (4μ)	(S)	$1/\sqrt{3}$
		(T)	3/16
		(U)	1/16

(A) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ Q
(B) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ S
(C) I $\rightarrow$ Q, II $\rightarrow$ S, III $\rightarrow$ R, IV $\rightarrow$ P
(D) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ R, IV $\rightarrow$ T

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5. Answer the following by appropriately matching the lists based on the information given in the paragraph.

A musical instrument is made using four different metal strings, 1,2,3 and 4 with mass per unit length μ, 2μ, 3μ and 4μ respectively. The instrument is played by vibrating the strings by varying the free length in between the range L_0 and $2L_0$. It is found that in string-1 (μ) at free length L_0 and tension T_0 the fundamental mode frequency is f_0.

List-I gives the above four strings while list-II lists the magnitude of some quantity. **[JEE-Advance-2019]**

	List-I		**List-II**
(I)	String-1(μ)	(P)	1
(II)	String-2 (2μ)	(Q)	1/2
(III)	String-3 (3μ)	(R)	$1/\sqrt{2}$
(IV)	String-4 (4μ)	(S)	$1/\sqrt{3}$
		(T)	3/16
		(U)	1/16

The length of the string 1,2,3 and 4 are kept fixed at $L_0, \frac{3L_0}{2}, \frac{5L_0}{4}$ and $\frac{7L_0}{4}$, respectively. Strings 1,2,3 and 4 are vibrated at their 1st, 3rd, 5th and 14th harmonics, respectively such that all the strings have same frequency. The correct match for the tension in the four strings in the units of T_0 will be.

(A) I $\rightarrow$ P, II$\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ U
(B) I $\rightarrow$ T, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ U
(C) I $\rightarrow$ P, II$\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ T
(D) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ T, IV $\rightarrow$ U

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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	(i)	(ii)	(iii)
Ans.	C	B	A	C	D	A		C	B	D
Que.	8	9	10	11	12	13	14	15	16	17
Ans.	B	A	B	A	C	A	D	B	C	D
Que.	18	19	20	21	22	23	24	25	26	27
Ans.	B	C	C	B	A	B	C	A	A	D
Que.	28	29	30	31						
Ans.	B	A	C	D						

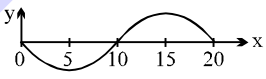
EXERCISE # O-2

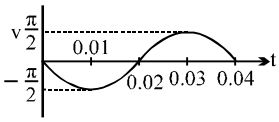
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	D	D	D	A	D	B	B	A
Que.	11	12	13	14	(i)	(ii)	(iii)	15	16	17
Ans.	C	A	D		C	C	B	A	A	A
Que.	18	19	20	21	22	23	24	25	26	27
Ans.	D	A	B	A	C	A	A,B,C,D	A,B,D	A,C,D	C,D
Que.	28	29	30	31	32	33	34	35		
Ans.	A,B,C	A,B,D	B,C,D	A,C,D	C,D	A	B	A		
Que.	36	(A)	(B)	(C)	(D)					
Ans.		P,R,S,T	S,T	Q,S,T	P,R,S,T					

EXERCISE # S

1. (a) $\lambda = 4m$, $f = \frac{1}{4}$ Hz, 1 m/s, (b) $\frac{3\pi}{2}$ mm/s, $\frac{3\pi^2}{4}$ mm/s^2, (c) $\frac{3\pi}{2}\times10^{-3}$ **2.** 1

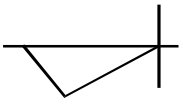
3. (a) DEF, (b) ABH, (c) CG, (D) AE **4.** $40\sqrt{2}$ **5.** 1.22 v **6.** 20 mm

7. 70 m/s **10.** (i) 

(ii) 25×10^{-6} W (iii) 

, solved

11. (a) negative x ; (b) $y = 4 \times 10^{-3} \sin 100\pi\left(3t + 0.5x + \frac{1}{400}\right)$ (x , y in meter) ; (c) $144\pi^2 \times 10^{-5}$ J

12. $\frac{25}{16}$ **13.** 49 mW **14.** – 0.35 cm. **15.** (a)  , (b) 2 cm/s

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16. $A_t = \frac{4}{3}A_i$, $A_r = \frac{1}{3}A_i$, $y_r = -\frac{0.04}{3}\sin(0.5\pi x + 100\pi t)$; $y_t = +\frac{0.16}{3}\sin(0.25\pi x - 100\pi t)$

17. 5 **18.** $10^9 N/m^2$ **19.** $E = \frac{A^2\pi^2 T}{4l}$ **20.** 500

21. 54.54 cm; 27.27; 18.18 cm **22.** x = 0.4m; 1 : $\sqrt{3}$

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	4	1	4	2	3	2	1	4	1
Que.	11	12	13	14						
Ans.	10	A	3	15						

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5
Ans.	B,C	A,C,D	A,D	A	A

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